

Exp 2: Implementation of Reinforcement Learning Framework and Markov Decision Process (MDP)

Aim

To implement a basic Reinforcement Learning framework based on the Markov Decision Process (MDP) and demonstrate the interaction between an agent and its environment using Python.

Algorithm

1. Start the program.
2. Define the **states, actions, rewards, transition probabilities, and discount factor** of the MDP.
3. Initialize the value of each state to zero.
4. Select an initial state.
5. Select an action for the current state.
6. Perform the action and observe the next state and reward.
7. Update the value of the current state using the Bellman equation:
8. Move the agent to the next state.
9. Repeat Steps 5–8 until the terminal state is reached.
10. Display the learned state values.
11. Stop.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

states = [0, 1, 2, 3]
actions = [0, 1]
rewards = {
    (0, 0): 0, (0, 1): 0,
    (1, 0): 0, (1, 1): 1,
    (2, 0): 0, (2, 1): 2
}
```

```

next_state = {
    (0, 0): 1, (0, 1): 2,
    (1, 0): 0, (1, 1): 3,
    (2, 0): 0, (2, 1): 3
}
gamma = 0.9
V = np.zeros(4)
for _ in range(100):
    new_V = V.copy()
    for s in states[:-1]:
        values = []
        for a in actions:
            ns = next_state[(s, a)]
            r = rewards[(s, a)]
            values.append(r + gamma * V[ns])
        new_V[s] = max(values)
    V = new_V
print("State Values:")
for s in states:
    print("State", s, ":", round(V[s], 3))
plt.figure(figsize=(7, 5))
plt.bar(states, V)
plt.xlabel("States")
plt.ylabel("Value")
plt.title("MDP State Value Function")
plt.xticks(states)
plt.grid(axis="y")
plt.show()

```

OUTPUT

State Values:

State 0 : 1.8

State 1 : 1.62

State 2 : 2.0

State 3 : 0.0

